

Element G: Construction of a Testable Prototype

The original plan to construct the design is as follows:

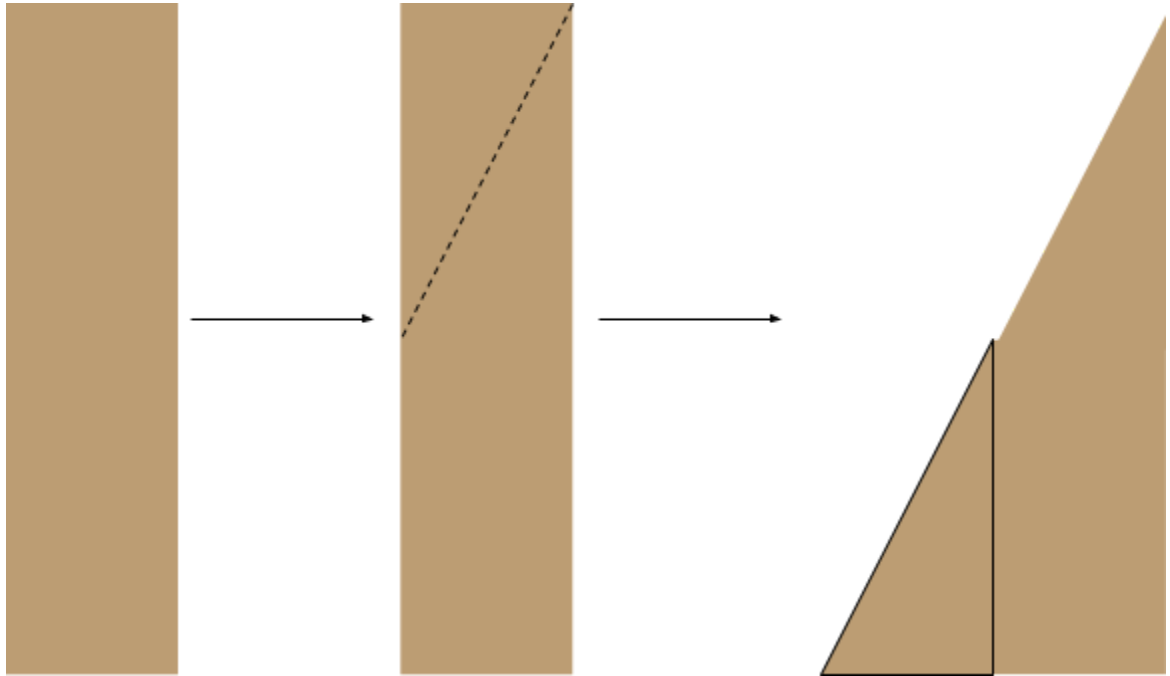
Procedures

1. Measure wood and draw a line from one corner to the other.
2. Cut wood along the line at the angle.
3. Prepare metal wire by straightening coat hangers.
4. Glue wire to wood
5. Paint

In actuality, there were many more steps in constructing the prototype. First, there were problems acquiring correctly sized wood. Since there was no wood that was thick enough, we elected to cut two pieces of wood and glue them together to get the required thickness. Wood glue was used, and it was left clamped to dry overnight. This was then cut again to get flat edges. Both cutting wood into two pieces and then cutting to flatten the edges was done by Mr Nusum, an engineering teacher at our school. The plan was to get the wood diagonally cut from corner to corner, but we were told that it was hard to do this. So, we thought of an alternative.

The alternative was to measure the midpoint of the side of the plank, and cut a triangle off of the plank. We would then combine this triangle to the main piece to create a bigger triangle. This way, we would have a thicker bottom side, creating more stability.

So with the alternative, the next diagonal cut to create the triangle, was done with a handsaw by members of our group. The wood was clamped to the table to be cut straightly. This cut left us with two pieces of wood, a smaller triangle shape, and a trapezoid shape, like a right triangle connected to a rectangle. We then glued the smaller triangle shape onto the trapezoid shape to create the slant of our bookend. We then used pliers to cut a hanger to get our wire as we bent it and straightened it into half of a square to assemble our platform where our books will be placed. To add more support, we cut up a binder to get a piece of cardboard that was the same size as our wire's shape to accommodate the support. We then glued those two pieces together and then attached it to our bookend with wood glue on the straight side of the bookend. After that, we decided to go with a flower design with vines on the side of the bookend and butterflies above the flowers.



Bookend Cutting Plan

Photos of Wire and Cardboard Piece Glued Attached to the Bookend

